

***Exploring Antibiofilm activity of bacteriocin
KHZ32 against
Healthcare-associated infectious pathogens***



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20UBE046

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APPROVAL SHEET

This report entitled “**Exploring Antibiofilm activity of bacteriocin KHZ32 against Healthcare-associated infectious pathogens**” by Chandradip Das is approved for the degree of B. Tech.

Examiners

Supervisor

Chairman

Date : _____

Place : _____

DECLARATION

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources that have thus not been properly cited or from whom proper permission has not been taken when needed.

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Chandradip Das

(Roll No.20UBE046)

Date: _____

CERTIFICATE

It is certified that the work contained in the report titled “**Exploring antibiofilm activity of bacteriocin KHZ32 against Healthcare-associated infectious pathogens**” Chandradip Das has been carried out under my supervision and that this work has not been submitted elsewhere for a degree.

Signature of Supervisor

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N.I.T. Agartala

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ABSTRACT

A bacterial strain isolated from Handia and identified as *Limosilactobacillus fermentum* KHZ32 was used in this study for its ability to produce antimicrobial compounds. The cell-free supernatant of *L. fermentum* KHZ32 showed clear zone of inhibition against two hospital-acquired pathogens *Staphylococcus aureus* and *Pseudomonas aeruginosa*. The bacteriocin present in cell-free supernatant of *L. fermentum* KHZ32 was precipitated by 80% w/v, ammonium sulfate. The protein concentration in the cell-free supernatant (CFS) and the dialyzed sample were found to be 0.292 mg/ml & 0.552 mg/ml respectively. The partially purified bacteriocin along with cell free supernatant of *L. fermentum* was used in antibiofilm assay by crystal violet staining using three indicator strains *S.aureus*, *P. aeruginosa* and *Escherichia coli*. When subjected to antibiofilm assay the bacteriocin KHZ329 (both crude and partially purified) showed significant reduction in the biofilm against the pathogens with a maximum reduction of almost 40% in the biofilm production of *Escherichia coli*. Additionally, the antibiofilm activity of bacteriocin from *L. fermentum* was validated by molecular docking, which displayed its strong binding affinity of -11.38kcal/mol and -11.406 kcal/mol, with ClfB and LecB, virulence factors from *S. aureus* and *P. aeruginosa* respectively.

Keywords: *Limosilactobacillus fermentum*, Bacteriocin, cell-free supernatant, antibiofilm assay.

1. Introduction

Healthcare-associated infections (HAIs) pose a significant challenge to public health worldwide, contributing to increased morbidity, mortality, and healthcare costs. These infections, acquired during the delivery of healthcare services in various settings such as hospitals, clinics, and long-term care facilities, are often associated with invasive procedures, immunocompromised patients, and the presence of multidrug-resistant pathogens. Among the diverse array of microbes implicated in HAIs, *Pseudomonas aeruginosa*, *Staphylococcus aureus*, and *Escherichia coli* stand out as prominent culprits (Smith et al., 2020).

Pseudomonas aeruginosa, a ubiquitous Gram-negative bacterium, is notorious for its ability to cause severe infections, particularly in immunocompromised individuals and those with underlying health conditions (Chatterjee et al., 2019). Its versatile metabolic capabilities and inherent resistance to many antibiotics make it a formidable foe in clinical settings. *Staphylococcus aureus*, a Gram-positive bacterium, is another common pathogen implicated in HAIs, ranging from superficial skin infections to life-threatening bloodstream infections (Rabin et al., 2021). The emergence of methicillin-resistant *Staphylococcus aureus* (MRSA) strains further complicates treatment options, highlighting the pressing need for novel therapeutic strategies. *Escherichia coli*, a Gram-negative bacterium commonly found in the gastrointestinal tract, can also cause HAIs, particularly urinary tract infections and surgical site infections (Beyrouthy et al., 2022).

One of the key mechanisms contributing to the virulence and persistence of these pathogens is the formation of biofilms. Biofilms are complex microbial communities encased within a self-produced extracellular matrix, providing protection against environmental stresses and host immune responses (Høiby et al., 2010). *Pseudomonas aeruginosa*, for instance, employs various virulence factors, including exopolysaccharides and extracellular DNA, to form robust biofilms on both biotic and abiotic surfaces (Rybtke et al., 2015). Similarly, *Staphylococcus aureus* produces a matrix composed of polysaccharides, proteins, and extracellular DNA, facilitating biofilm formation on medical devices and host tissues (Foulston et al., 2014). *Escherichia coli*, although primarily associated with planktonic growth, can also form biofilms on mucosal surfaces and indwelling medical devices, contributing to recurrent infections and treatment failure (Römling et al., 2014).

The presence of biofilms poses significant challenges in the treatment of HAIs. Biofilm-associated bacteria exhibit increased resistance to antibiotics, disinfectants, and host immune defenses compared to their planktonic counterparts (Bjarnsholt et al., 2013). The dense matrix limits the penetration of antimicrobial agents, while the physiological heterogeneity within the biofilm fosters the emergence of persister cells, which are metabolically dormant and highly tolerant to antimicrobial treatment (Santajit and Indrawattana, 2016). Moreover, the upregulation of efflux pumps and the formation of small colony variants further contribute to antibiotic resistance and treatment failure (Fisher et al., 2018).

In light of the challenges posed by biofilm-associated infections, there is growing interest in exploring alternative therapeutic approaches. Bacteriocins, antimicrobial peptides produced by bacteria to inhibit the growth of closely related strains, have emerged as promising candidates for combating biofilm formation and dispersal (Yang et al., 2020). Unlike conventional antibiotics, bacteriocins exhibit narrow-spectrum activity, targeting specific bacterial species or strains without disrupting the commensal microbiota (Tagg et al., 1976). Furthermore, bacteriocins can disrupt biofilms by targeting key components of the extracellular matrix or by inhibiting quorum sensing, a process critical for coordinating biofilm formation and virulence factor expression (Cotter et al., 2013). For instance, bacteriocins such as nisin have been shown to inhibit biofilm formation by *Staphylococcus aureus* and *Escherichia coli* through mechanisms involving cell membrane disruption and interference with cell-cell communication (Field et al., 2015; Khelissa et al., 2021). Similarly, pyocins produced by *Pseudomonas aeruginosa* have demonstrated efficacy in preventing biofilm formation and eradicating pre-existing biofilms in both in vitro and in vivo models (Michalska et al., 2019). In conclusion, HAIs remain a significant threat to patient safety and healthcare delivery, with multidrug-resistant pathogens and biofilm formation exacerbating the challenge. Understanding the mechanisms underlying biofilm formation by key pathogens such as *Pseudomonas aeruginosa*, *Staphylococcus aureus*, and *Escherichia coli* is crucial for developing effective therapeutic strategies. The potential of bacteriocins as antibiofilm agents offers new hope for mitigating the impact of HAIs and improving patient outcomes in clinical settings.

2. Materials and Methods

2.1 Chemical reagents

The components of MRS media and Nutrient Broth media, TSB media were purchased from Hi-Media India Pvt., Ltd. All the chemicals required for Antibiofilm Assay, protein estimation and buffer preparations for dialysis were of high purity grade and purchased from Hi-Media Pvt. Ltd., India.

2.2 Media preparation

MRS media contained (percentage weight/volume) Peptone: 10%, Beef extract: 10%, Yeast extract: 5%, Dextrose (glucose): 20%, Dipotassium hydrogen phosphate: 2%, Sodium acetate: 0.5%, Ammonium citrate: 2%, Magnesium sulfate: 0.1%, Manganese sulfate: 0.05%, Tween 80: 1%. Tryptic Soy Broth (TSB) media contained Pancreatic Digest of Casein: 15%, Papaic Digest of Soybean Meal: 5%, Sodium Chloride: 0.5%, Dipotassium Phosphate: 0.5%, Glucose: 1%, Distilled Water: Balance to 100%. Both media were first autoclaved at 121 °C for 20 min and then cooled before the inoculation.

2.3 Extraction of cell-free supernatant (CFS) from producing strain

The bacterial strain was inoculated in MRS media (5ml) and then incubated for 24 hours in the incubator at 37°C inside the incubator cum shaker. After 24 hours, centrifugation was done at 10000 rpm for 15mins at 4°C. The cell-free supernatant (4.1ml) was considered as crude bacteriocin. The pH of the cell-free supernatant was measured using pH meter (pH=6.5). The cell-free supernatant was further used for antimicrobial assay and antibiofilm assay.

2.4 Purification of Bacteriocin KHZ from CFS

The bacterial strain was inoculated in MRS media (100ml) and incubated for 24 hrs at 37°C. The cell-free supernatant was obtained via centrifugation at 10000rpm at 4°C. The CFS was further partially purified by ammonium sulfate precipitation (80% saturation) followed by dialysis in 50 mM sodium acetate buffer (pH 6.5). The dialyzed sample was further used for antimicrobial and antibiofilm assay.

2.5 Antimicrobial activity test using agar well diffusion

Both crude and partially purified bacteriocin was used for antimicrobial assay by agar well diffusion method using *Staphylococcus aureus* MTCC-7814 as indicator strain. *Staphylococcus*

aureus MTCC-7814 was inoculated in MRS (5ml) and incubated for 24 hrs at 37°C. The bacteria cells were then separately added (50µl) to NB Agar (2%) media. After the solidification the wells were made on each plate. Crude and Dialysed sample (70µl each) were separately given in the wells. Ampicillin(5mg/ml) was used as standard and also given in a separate well for comparison of antagonistic activity. The plate was incubated at 37°C for 18-20 h. After the incubation, the zone of inhibition around the well was measured

2.6 Determination of protein concentration using BSA

The protein concentration of the cell free supernatant of the isolate was estimated following the Bradford method of using BSA as standard. The protein estimation was carried out by adding 500µl of cell free supernatant of isolate to 500µl of sodium acetate buffer (pH 6.5) and incubated at room temperature for 15 min. After the incubation, 1000µl of Bradford reagent was added to the reaction mixture and absorbance (A₅₉₅) of colour developed was measured at 595 nm on a UV-visible spectrophotometer.

$$\text{Protein concentration (mg/ml)} = \frac{A_{590} \times C \times V}{v}$$

ΔA_{590} = Optical Density (OD) change at 590 nm.

C = 1 OD equivalent BSA concentration (mg/ml) from standard plot.

V = volume of the reaction mixture (ml).

v = volume of protein sample (ml).

2.7 Calculation of Activity unit (AU)

The antibacterial activity results of the fractions which were expressed in arbitrary units (AU/ml) using a serial dilution method in which five wells were drilled. Well1 (D0) having 100µl crude filter sterilized CFS, Well2(D1) having 100µl D0 and 100µl sterile dH2O, Well3(D2) having 100µl D1 and 100µl sterile dH2O, Well4(D3) having 100µl D2 and 100µl sterile dH2O and Well5(D4) having 100 µl D3 and 100µl sterile dH2O. The arbitrary unit of bacteriocin was calculated according to the formula

$$AU = a^b \times 10$$

where a= dilution factor

b= last dilution which shows clear zone of inhibition of 2mm

2.8 Specific activity calculation

The specific activity (U/mg) of the cell free supernatant was calculated using the following equation;

$$\text{Specific activity (U/mg)} = \frac{\text{Enzyme Activity (U/ml)}}{\text{Protein Concentration (mg/ml)}}$$

2.9 Antibiofilm Assay

2.9.1 Conformation of biofilm formation

To confirm the biofilm formation in *Pseudomonas aeruginosa* MTCC 7814, *Staphylococcus aureus* MTCC 87 & *Escherichia coli* MTCC 1302 were grown in TSB media (3 %) with agar (1.5%) and Congo red (0.08%). The bacteria were inoculated (streaking method) on TSB agar plate and incubated for 48 hrs. The colonies were observed to confirm the biofilm formation. *Limosilactobacillus fermentum* CFW6 (a biofilm producing bacteria) & *Sporosarcina contaminans* AKMT9 (with no biofilm production) were also to compare the colonies.

2.9.2 Antibiofilm Assay of Bacteriocin KHZ32

The effects of Bacteriocin KZ32(CFS & Dialyzed form) on biofilm formation by *P. aeruginosa* MTCC 7814, *S. aureus* MTCC 87 & *E. coli* MTCC 1302 were measured in a 96-well microplate and stained with crystal violet. In triplicates, exponential phase bacterial suspensions (OD₆₀₀ = 0.4, 200µL) were put to a 96 well microplate with 100µL CFS and Dialysed sample from *Limosilactobacillus fermentum* KHZ32 at concentration of 0.362mg/ml & 0.552mg/ml respectively. After 48 h of incubation at 37 °C, well contents were discarded and biofilm was washed three times with distilled water. Afterward that wells were fixed with 100L methanol for 20 min, dried at 25 °C, and stained with 100 µL 1.0% (w/v) crystal violet for 15 min. The dye bound biofilm was dissolved in 250µL of 95% ethanol and absorbance was taken at 595 nm. The antibiofilm activity was represented as

% Biofilm reduction = (1 - (As/Ac)) x 100 where,

A = absorbance at 595 nm.

The indicator strains termed as control (Ac) and CFS and Dialyzed sample treated indicator strains termed as samples (As).

2.10 In silico analysis of antibiofilm activity

2.10.1 Target and Ligand Preparation

Total five genes responsible for biofilm-production and quorum sensing in *Pseudomonas aeruginosa* (PelA, PelD, LecB, LasR, LasI) and *Staphylococcus aureus* (IcaR, Fib, ClfB, SasG, Cna) were curated from literatures and research papers. Their structures were predicted using homology modelling in SWISS MODEL (<https://swissmodel.expasy.org/>) and 3D models were constructed based on the templates (<2 Armstrong) obtained in homology modelling and downloaded in PDB format. The 3D structures were further refined by adding polar hydrogen bonds and removing of any ions or water particles in Biovia Discover studio. The structure of bacteriocin (ligand) was downloaded from NCBI in fasta format. The 3D structure of the ligand was constructed using AlphaFold. The 3D structure of the ligand was also visualized in Discovery Studio.

2.10.2 Molecular docking and Molecular dynamics simulations

Molecular docking was performed between the five target genes of the *P.aeruginosa* and also the five target genes of *S.aureus* with the fermentum as ligand using AutoDock Vina. Before the docking, the ligand and receptors were prepared in pdbqt format and grid box were set in AutoDock. During the docking process, a maximum of eight conformers were considered for each ligand. The most favourable binding pose was chosen based on the lowest free energy of binding (ΔG) and the molecular interactions between the ligands and receptors were analysed using Discovery Studio.

3. Flow diagram of methods

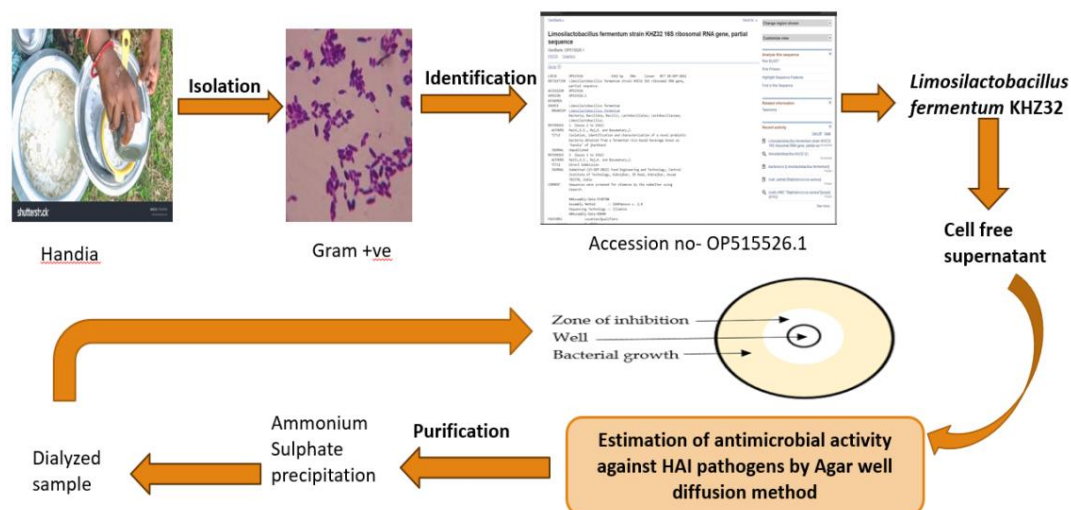


Fig. 1. The work flow of purification & antimicrobial assay of Bacteriocin KHZ32

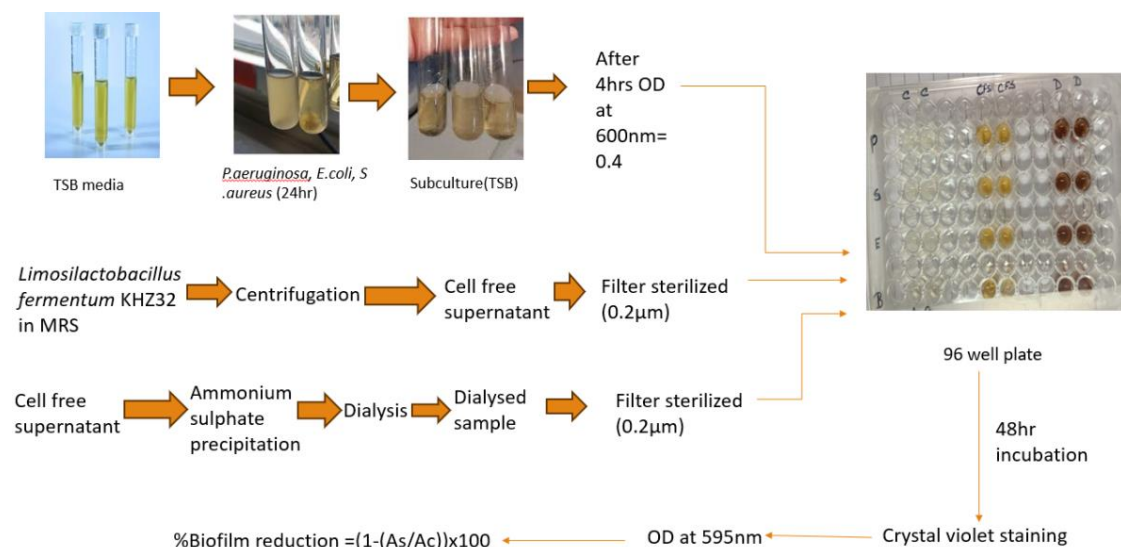


Fig. 2. Protocol for Antibiofilm assay using crystal violet

Limosilactobacillus fermentum KHZ32 was grown and then centrifuged to obtain the cell-free supernatant (CFS). The CFS was further purified to get the Dialyzed sample and then used in antimicrobial activity test as shown in Fig1. The CFS and Dialyzed sample were then used in antibiofilm assay as shown Fig 2.

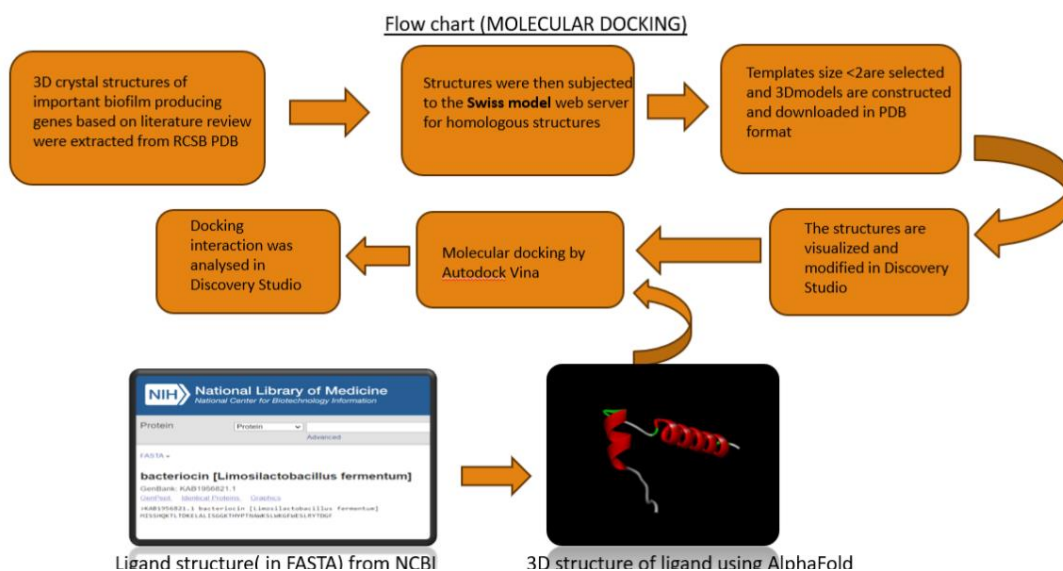


Fig. 3. In silico analysis of ligand against target receptors

The ligand structure was downloaded in FASTA and 3D model was constructed using Homology Modelling. In the similar way the structures of the receptors (genes responsible for biofilm formation in *S. aureus*, *P. aeruginosa*, & *E. coli*) were downloaded from **RCSB PDB** (<https://www.rcsb.org/>) then subjected Homology Modelling and then the 3D structures were constructed as shown in Fig3.

4. Results and Discussion

4.1 Antimicrobial assay

Antimicrobial assay was performed using agar well diffusion method. The indicator strain used for this was *S. aureus*. The antimicrobial activity of bacteriocin KHZ32 was clearly observed (as shown in Fig.3). Fig. (A) illustrates the agar plate of *Staphylococcus aureus* (with CFS & Ampicillin). After 24 hours of incubation, there was a distinct inhibition ring observed in both well 1 (Ampicillin) and well 2 (CFS), with diameters of 32 mm and 20 mm, respectively. Similarly, Fig. (B) showcases the agar plate of *Staphylococcus aureus* (with Dialyzed sample & Ampicillin). Following 24 hours of incubation, clear inhibition rings were evident in both well 1 (Ampicillin) and well 2 (Dialyzed sample), measuring 36 mm and 30 mm in diameter, respectively. Its evident that Bacteriocin KHZ32(in CFS & Dialyzed form) has antimicrobial activity against *S. aureus*.

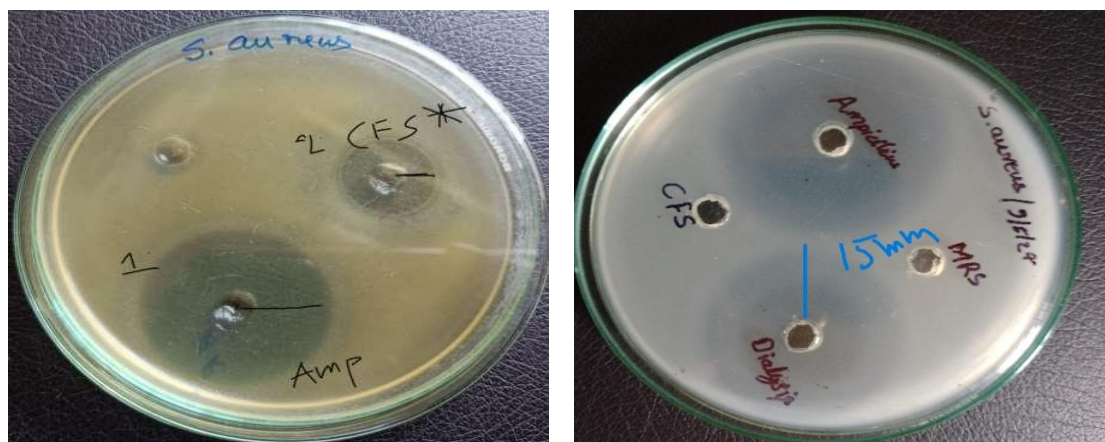


Fig. 4. (A) Ampicillin, CFS against *S. aureus* (B)Ampicillin, Dialysed sample against *S. aureus*

4.2 Protein concentration, Activity Unit and Specific Activity Calculation

4.2.1 Protein concentration

In order to calculate the protein concentration in cell free supernatant, the BSA standard curve analysis is done at pH 6.5 . The standard curve of BSA is shown in Fig. 5.

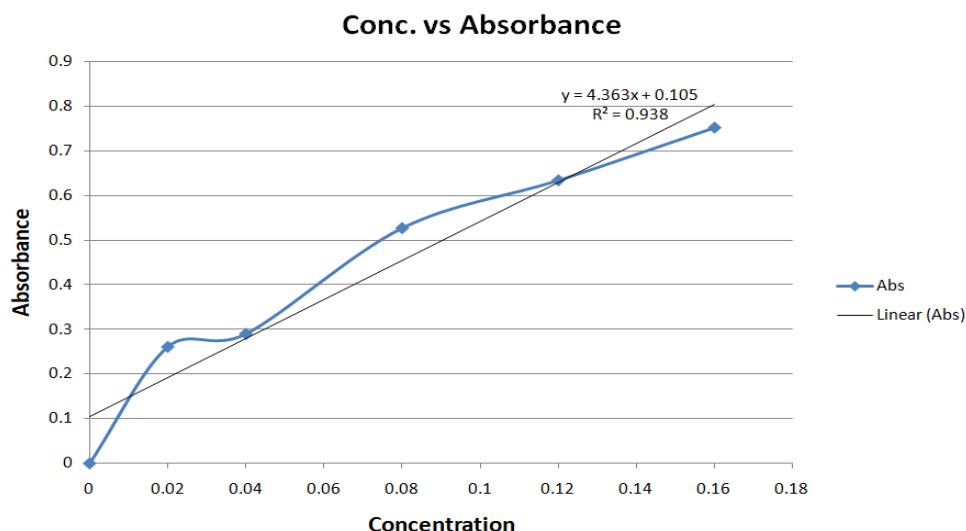


Fig. 5. BSA Standard curve at pH 6.5

From the above equation, the protein concentration of CFS from *L. fermentum* KHZ32 was found to be **0.292 mg/ml** and in Dialyzed sample it was **0.552mg/ml**

4.2.2 Activity unit Calculation

For the calculation of AU/ml, the filter sterilized CFS & Dialyzed sample were serially 2 fold diluted.

In case of *S.aureus* serial dilution (CFS) upto D3 showed clear zone of inhibition upto 2 mm. Therefore; $AU = 2^3 \times 100 = 800 \text{ AU/ml}$

In case of *S.aureus* serial dilution (Dialyzed) upto D5 showed clear zone of inhibition upto 2 mm. Therefore; $AU = 2^5 \times 100 = 3200 \text{ AU/ml}$

The results obtained from both the CFS and dialyzed samples showed differences observed in their activity units. Dialyzed sample activity unit was 4 times higher than CFS sample. The significance of these differences in terms of the presence or absence of certain molecules or compounds in the dialyzed samples compared to the CFS.



(A) Activity unit calculation *S.aureus*(CFS) (B) Activity unit calculation *S.aureus*(DIA)

Fig 6: Agar well plate for activity unit calculation of CFS and Dialysed sample against *S.aureus*

4.2.3 Specific activity calculation

Specific Activity of CFS:

$$\begin{aligned}\text{Specific Activity} &= \frac{\text{AU/ml}}{\text{mg/ml}} \\ &= 800\text{AU/ml} / 0.292 \text{ mg/ml} \\ &= \mathbf{2739.7 \text{ AU/mg}}\end{aligned}$$

Specific Activity of Dialyzed sample:

$$\begin{aligned}\text{Specific Activity} &= \frac{\text{AU/ml}}{\text{mg/ml}} \\ &= 3200\text{AU/ml} / 0.552 \text{ mg/ml} \\ &= \mathbf{5797.1\text{AU/mg}}\end{aligned}$$

Table 1. Optimum Activity of bacteriocin KHZ32 against *S.aureus*

Sample	Concentration mg/ml	Activity AU/ml	Specific activity AU/mg
CFS	0.292	800	2739.7
Dialyzed	0.552	3200	5797.1

4.3 Antibiofilm assay

4.3.1 Confirmation of biofilm formation

To confirm the biofilm formation the colonies were observed after 48 hrs of incubation.

(A), (B), (C), shows clear colonies with thick slime layer. This indicates the presence of biofilm (with low EPS). Hence *P. aeruginosa*, *S. aureus* & *E. coli* are biofilm positive.

(C) shows clear colonies with black slime which indicates presence of biofilm (with high EPS).

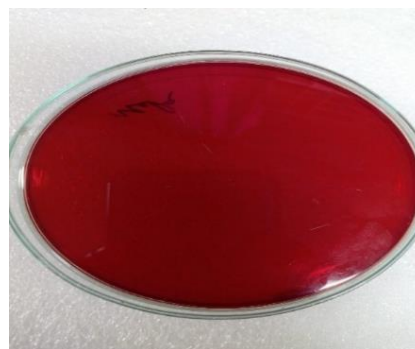
(D) there are only small micro colonies with no slime layer which indicates absence of biofilm.



(A) *S. aureus* MTCC 87

(B) *P. aeruginosa* MTCC 7814

(C) *E. coli* MTCC 1302



(C) *S. contaminans* AKM19

(D): *L. fermentum* CFW6

Fig.7 : Confirmation of biofilm formation

4.3.1 Antibiofilm assay of Bacteriocin KHZ32 in CFS and Dialyzed form

After 48 hrs incubation and subsequent Crystal violet staining OD were taken at 595nm to calculate the biofilm reduction %. The given table shows the absorbance data.

$$\% \text{ Biofilm reduction} = (1 - (As/Ac)) \times 100$$

PA= *Pseudomonas aeruginosa*

SA= *Staphylococcus aureus*

EC= *Escherichia coli*

CFS= 100 μ L filter sterilized CFS of *L. fermentum* KHZ32 (0.362mg/ml)

Dia= 100 μ L filter sterilized dialyzed sample of *L. fermentum* KHZ32 (0.552mg/ml)

Average biofilm reduction of 27.27%, 29.50% & 37.15% in case of *Pseudomonas aeruginosa*, *Staphylococcus aureus* & *Escherichia coli* respectively. Dialyzed sample showed a biofilm reduction of 30.12%, 31.89% & 39.94% in case *Pseudomonas aeruginosa*, *Staphylococcus aureus* & *Escherichia coli* respectively. It is evident that Bacteriocin KHZ32 has antibiofilm activity against *S.aureus*, *P.aeruginosa*, *E.coli* in CFS and Dialyzed form & in the dialysed sample the antibiofilm activity was slightly increased and the biofilm reduction % was maximum for *Escherichia coli*(39.94%).

Table 2: Absorbance data at 595nm

	OD1		OD2		OD avg
TSB Only (Blank)	0.196		0.114		0.155
PA Only (Control)	0.277		0.284		0.281
SA Only (Control)	0.168		0.171		0.170
EC Only (Control)	0.164		0.159		0.162
Sample	OD1	% Reduction	OD2	% Reduction	Average (%)
CFS+PA	0.210	25.134	0.198	29.412	27.27
CFS+SA	0.118	30.383	0.121	28.614	29.50
CFS+EC	0.108	33.127	0.095	41.176	37.15
Dia+PA	0.191	31.907	0.201	28.342	30.12
Dia+SA	0.108	33.127	0.112	30.650	31.89
Dia+EC	0.091	43.653	0.103	36.223	39.94

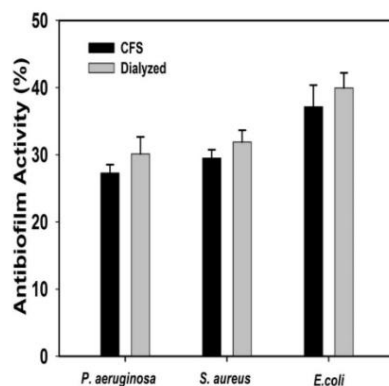


Figure 8: Bar graph of antibiofilm assay

4.4 Molecular docking

4.4.1 Ligand and Receptor structure prediction

The 3D structure of our ligand was predicted using Alpha Fold which provided best 5 models based on higher sequence coverage with our FASTA sequence of *L. fermentum* which is obtained from NCBI database under the accession no KAB1956821.1

```
>KAB1956821.1 bacteriocin [Limosilactobacillus fermentum]
MISSHQKTLTDKELALISGGKTHYPTNAWKSLWKGFWESLRYTDGF
```

Figure 9: Fasta sequence of our *L. fermentum* Ligand

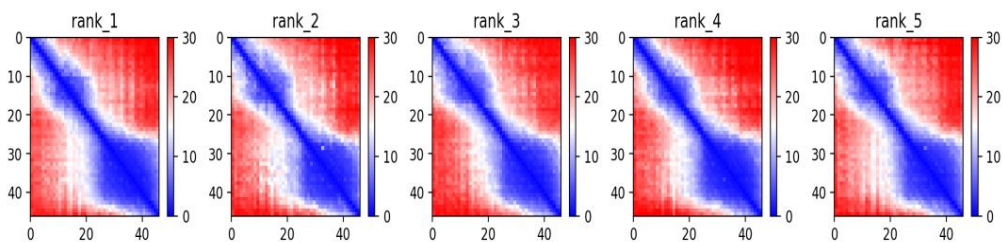


Figure 10: Best 5 predicted models in Alpha fold

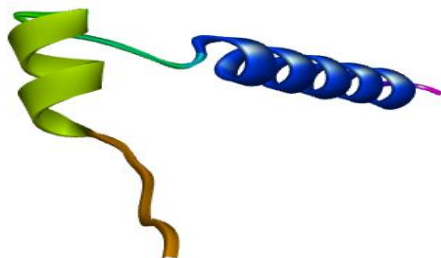
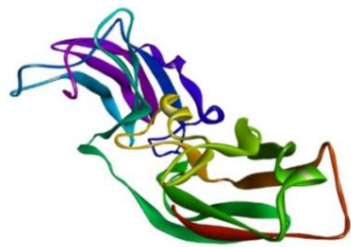


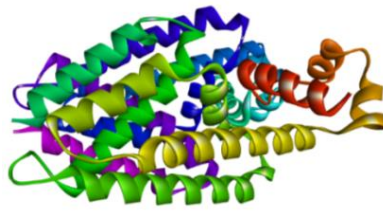
Figure 11: 3D Structure of the ligand predicted by Alphafold

In the same way as above, templates less than 2 Angstrom were selected after modelling, for best docking results and the 3D structures were downloaded in PDB format and subjected to molecular docking.

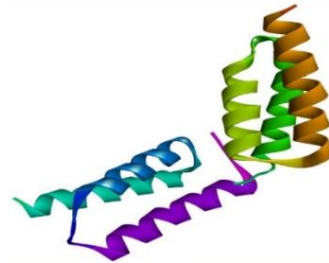
The Lamarckian Genetic Algorithm and **AutoDock Vina** were used to execute molecular docking studies with ten target receptor proteins as those involved in biofilm formation and quorum sensing. The most favourable binding pose was chosen based on the lowest free energy of binding (ΔG) and the molecular interactions between the ligands and receptors were analysed using Discovery Studio. Molecular docking, an integral part of structure-based drug design, allows the virtual selection of probable drug candidate molecules by evaluating their binding affinity with the target receptors.



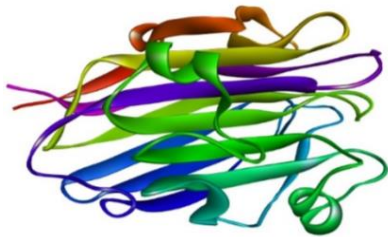
(A) ClfB



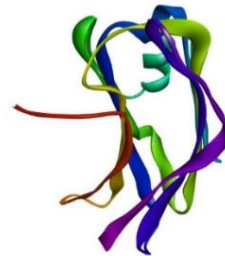
(B) IcaR



(C) Fib



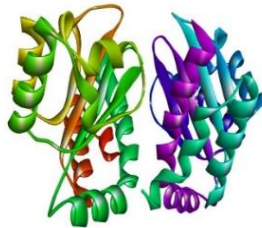
(D) SasG



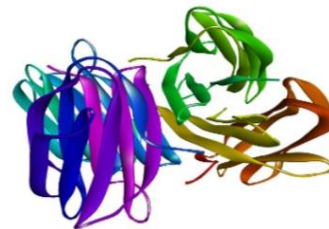
(E) Cna

Fig.12: 3D models of *S.aureus* (virulence factor)

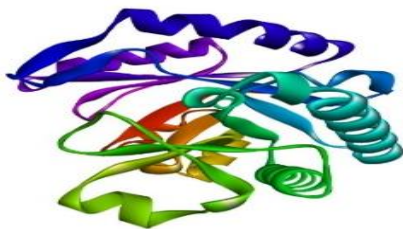
(A) LasI



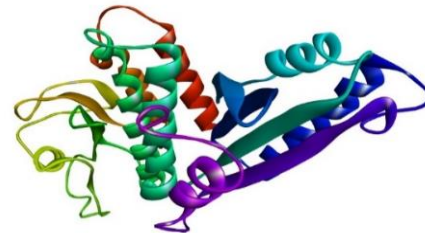
(B) LasR



(C) LecB



(D) PelA



(E) Pel D

Fig.13: 3D models of *P. aeruginosa*

Table 3: Target receptors and their docking scores with ligand(*S.aureus*)

SL NO	Target Receptors(<i>S.aureus</i>)	References	Binding Affinity (-kcal/mol)
1	IcaR (transcription regulation in the process of biofilm formation)	Dan Yu, Liping Zhao et al., 2012	-8.787
2	Fib (helps to recognize surface-fibrinogen binding proteins)	Nathan K Archer et al., 2011	-9.248
3	SasG(The surface protein promotes biofilm formation during the accumulation phase)	TJ Foster et al., 2016	-9.407
4	ClfB (accumulates on the bacterial surface and mediates biofilm formation)	Nabil M. Abraham et al., 2012	-11.38
5	Cna(contributes to the initial attachment of <i>S. aureus</i> cells to surfaces. Once attached, the bacteria can begin to form biofilm.)	Philippe Herman Bausier et al .2016	-9.133

ClfB that helps in the formation of biofilm in *Staphylococcus aureus* showed the highest docking score indicating significant and high binding affinity of the ligand

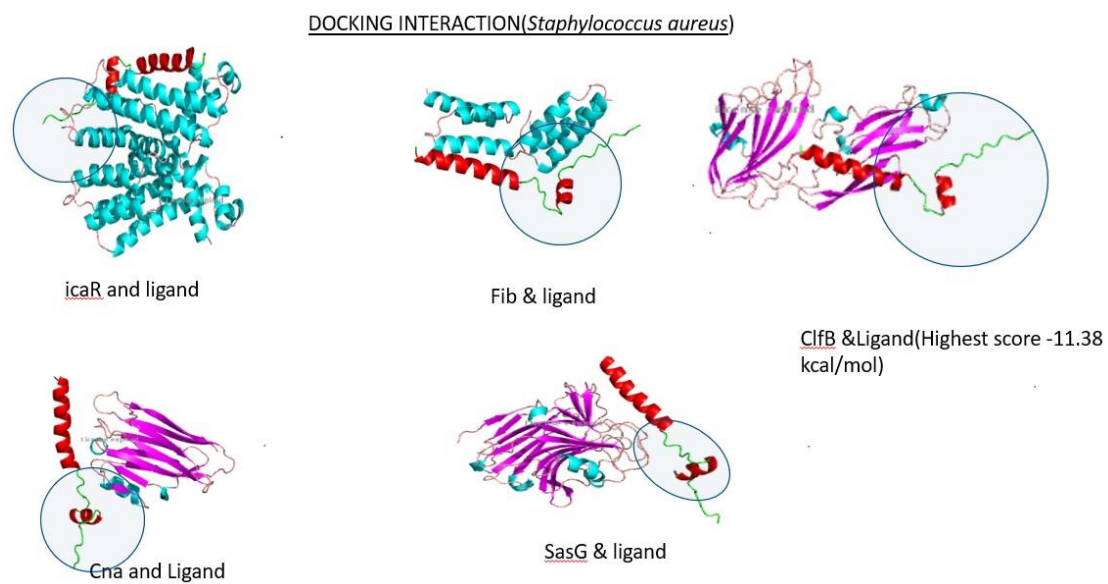
Table 4 : Target receptors and their docking scores with ligand(*P. aeruginosa*)

SL NO	Target Receptors(<i>P. aeruginosa</i>)	References	Binding Affinity (-kcal/mol)
1	PelA(involved in the biosynthesis of Pel polysaccharide of extracellular matrix)	Kelly M. Colvin et al., 2011	-9.984
2	PelD(biosynthesis of Pel polysaccharide, which is a crucial component in the formation of biofilm)	Gregory B. Whitfield et al., 2020	-10.798
3	LasR(positively regulates biofilm generation)	Tsiry Rasamiravaka et al., 2015	-11.376
4	LasI(regulates the biosynthesis of biofilm)	Teresa R. De Kievit et al., 2001	-10.107
5	LecB (facilitates bacterial adhesion and aggregation, thus enhancing the stability and structural integrity of the biofilm)	Roman Sommer et al., 2018	-11.406

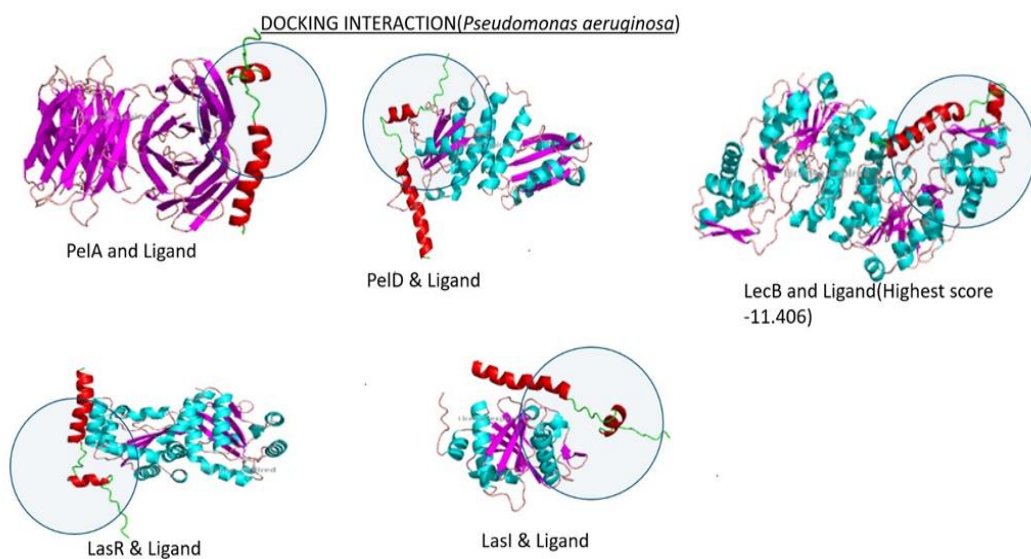
LecB showed the highest docking score which indicates high binding affinity of the ligand .

4.4.2 Ligand and receptor interaction

The commendable docking scores between Ligand and the receptors imply significant binding affinity of the ligand towards the receptors. This implies promising inhibition potential of the ligand towards the receptors thus leading to inhibition of biofilm formation.



A



B

Figure 14: Docking interaction between (A) *S. aureus* biofilm forming genes and ligand (B) *P. aeruginosa* biofilm producing genes and ligand

Ligand showed significant and high binding affinity towards the binding site.

5. Conclusion

The results of the biofilm assay showed commendable inhibition potential of Bacteriocin KHZ32 in biofilm production/positive antibiofilm activity of Bacteriocin KHZ32 against *Pseudomonas aeruginosa*, *Staphylococcus aureus* & *Escherichia coli* (the most common Healthcare Associated Infection Pathogens). The highest biofilm reduction % was observed against *Escherichia coli* (almost 40%). The In-silico studies further implied to significant binding affinity of the Ligand (Bacteriocin KHZ32) towards the target receptors (important genes necessary for biofilm production in 3 above mentioned bacteria) with highest binding affinity of **11.406 Kcal/mol** with **LecB** (a biofilm producing gene of *Pseudomonas aeruginosa*). This indicates inhibition of the receptors and consequently reduction in the biofilm formation. Moreover, bacteriocin KHZ32 showed high antimicrobial activity and specific activity against *Staphylococcus aureus* in crude (CFS) and purified form (Dialyzed form). Hence, Bacteriocin KHZ32 can be used a potential drug or adjuvant during the treatment of Healthcare Associated Infections especially in the treatment of HAIs caused by *S.aureus*, *P.aeruginosa*, *E.coli*.

6. Future prospects

Novel Therapeutic Approach: Bacteriocins are peptides produced by bacteria to inhibit the growth of competing bacteria. KHZ32, being a bacteriocin, represents a novel therapeutic approach to combatting healthcare-associated infections (HAIs). Unlike traditional antibiotics, which are losing efficacy due to resistance, bacteriocins like KHZ32 may offer a new avenue for treatment.

Biofilm Disruption: Biofilms are communities of bacteria encased in a protective matrix, making them highly resistant to antibiotics and immune responses. The ability of KHZ32 to disrupt biofilms is particularly promising as biofilm-related infections are notoriously difficult to treat and are a significant contributor to the persistence of HAIs.

Reducing Antibiotic Resistance: Given the rise of antibiotic resistance, there's an urgent need for alternative antimicrobial agents. Bacteriocins like KHZ32 represent a potential solution as they may have different mechanisms of action compared to conventional antibiotics, making them effective against resistant pathogens.

Synergistic Therapies: Research can explore the potential synergistic effects of bacteriocin KHZ32 with existing antibiotics or other antimicrobial agents. Combining bacteriocin KHZ32 with conventional therapies may enhance the efficacy of treatment regimens and reduce the emergence of antibiotic resistance.

7. References

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